

Mapping Exposures and Cumulative Risk of Hazardous Air Pollutants in the United States

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Hazardous Air Pollutants (HAPs)

- **188 compounds** designated by EPA since 1990
- EPA regulates HAPs emissions, but unlike criteria pollutants there are **no ambient concentration standards** regarding HAPs exposure
- The HAPs database combines **monitoring data collected by multitude entities**: the EPA, other federal, state, local, and tribal agencies, and other academic, community, and short-term studies
- Extensive toxicology data on HAPs **adverse effects**: cancer, reproductive effects, and birth defects, among others
- **Need for more exposure and epidemiology studies** to understand public health impacts of HAPs

Initial study: 4 HAPs, 2021

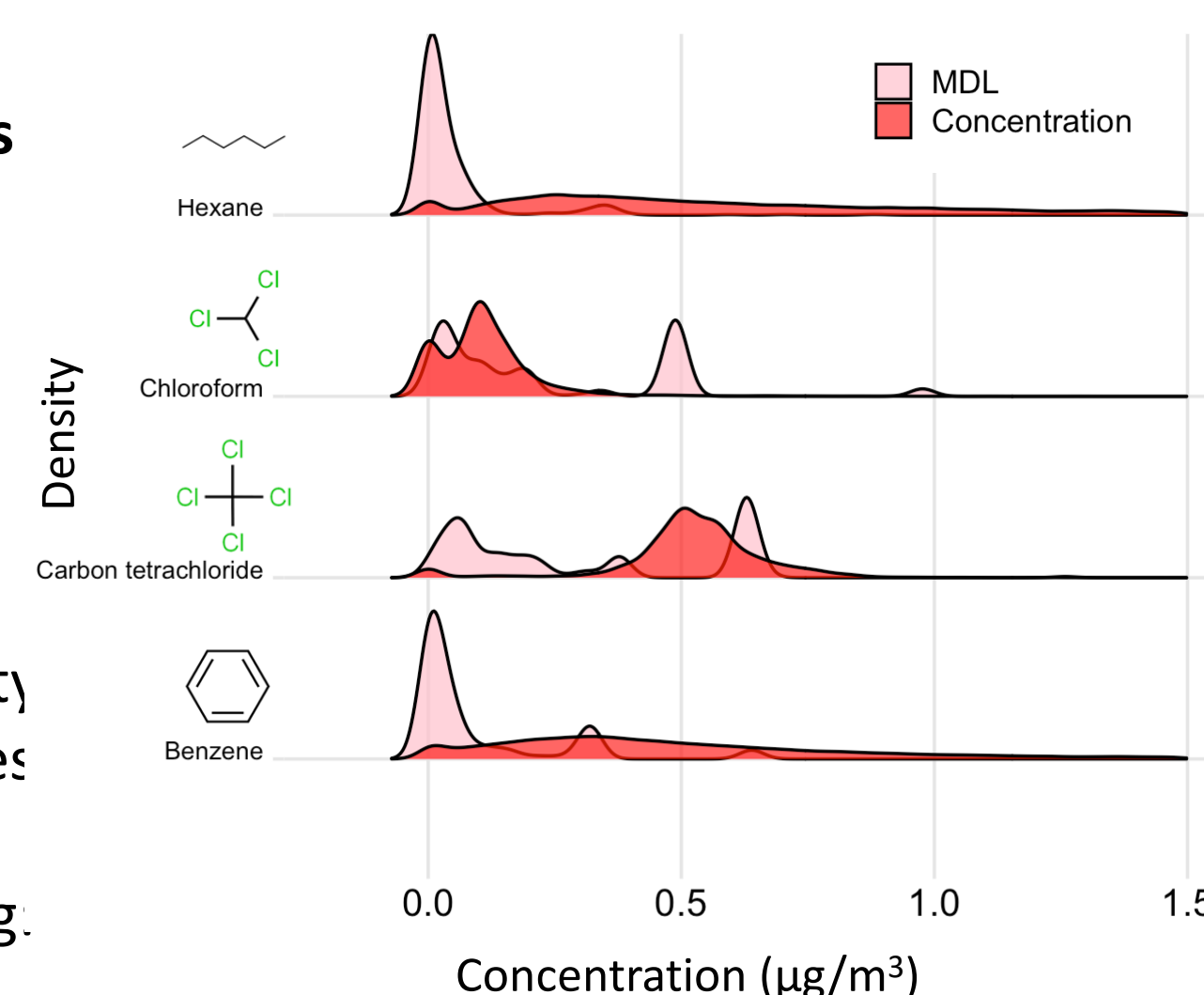
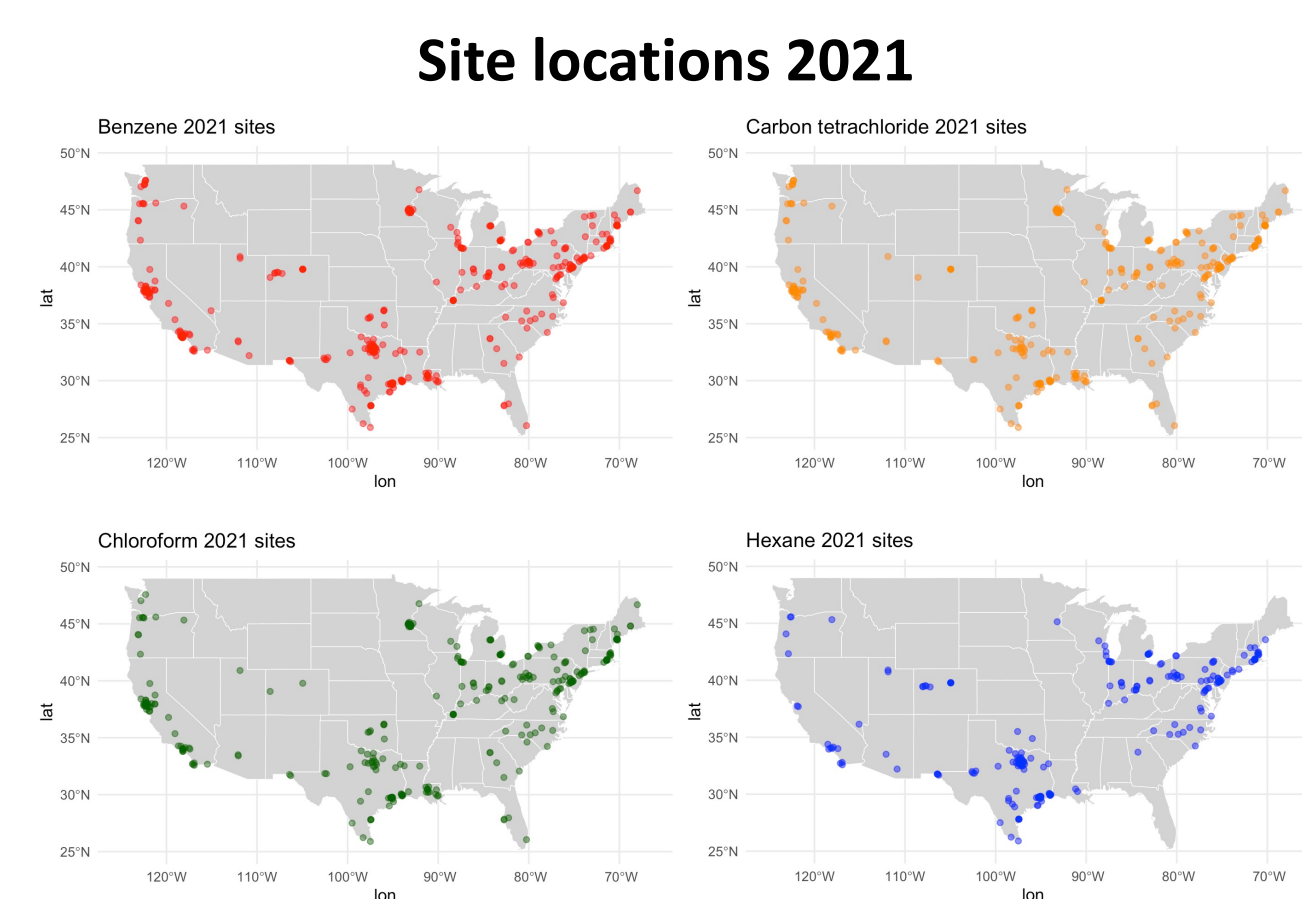
Pollutant	2021 Sites	Mean MDL	% Non-Detects	% Below MDL	Mean days/site
Benzene	282	0.21	2.2	8.9	97
Carbon tetrachloride	226	0.30	3.2	25.8	46
Chloroform	225	0.24	21.0	32.8	46
Hexane	165	0.05	3.6	2.1	119

Selection criteria:

- Included in Tox21 database
- 33% or less non-detects
- 150 or more sites
- ~50% or more measurements above Min. Detection Limit (MDL)

Dataset challenges:

1. **Sparsity** in both space and time: few sites, most not sampled daily
2. **Inhomogeneous sampling methods** and frequency: data combined from different agencies
3. **Censored data** at sub-daily measurement scales: need special considerations for daily average computation (censored MLE)
4. **Diversity of chemical classes**: variety of effects of chemistry, physics, sources and health effects
5. **Computationally complex** modeling: simultaneous fitting for multiple HAPs and a large number of covariates



References:

Bair E, Kidd B, Wimberley E, Mav D, Messier K (2024). PrestoGP: Penalized Regression for Spatio-Temporal Outcomes via Gaussian Processes. R package version 0.2.0.9036. <https://niehs.github.io/PrestoGP/>.
Climate and Health Outcomes Research Data System (CHORDS). <https://www.niehs.nih.gov/research/programs/chords>.
Landau WM (2021). "The targets R package: a dynamic Make-like function-oriented pipeline toolkit for reproducibility and high-performance computing." *Journal of Open Source Software*, 6(57), 2959. <https://doi.org/10.21105/joss.02959>.
Manware M, Song I, Marques E, Alifa Kassien M, Messier K (2024). amadeus: Accessing and Analyzing Large-Scale Environmental Data. R package version 1.2.0. <https://niehs.github.io/amadeus/>.
Messier KP, Katzfuss M. Scalable penalized spatiotemporal land-use regression for ground-level nitrogen dioxide. *Ann Appl Stat*. 2021 Jun;15(2):688-710. doi: 10.1214/20-aas1422. Epub 2021 Jul 12. PMID: 35069963; PMCID: PMC8774268.

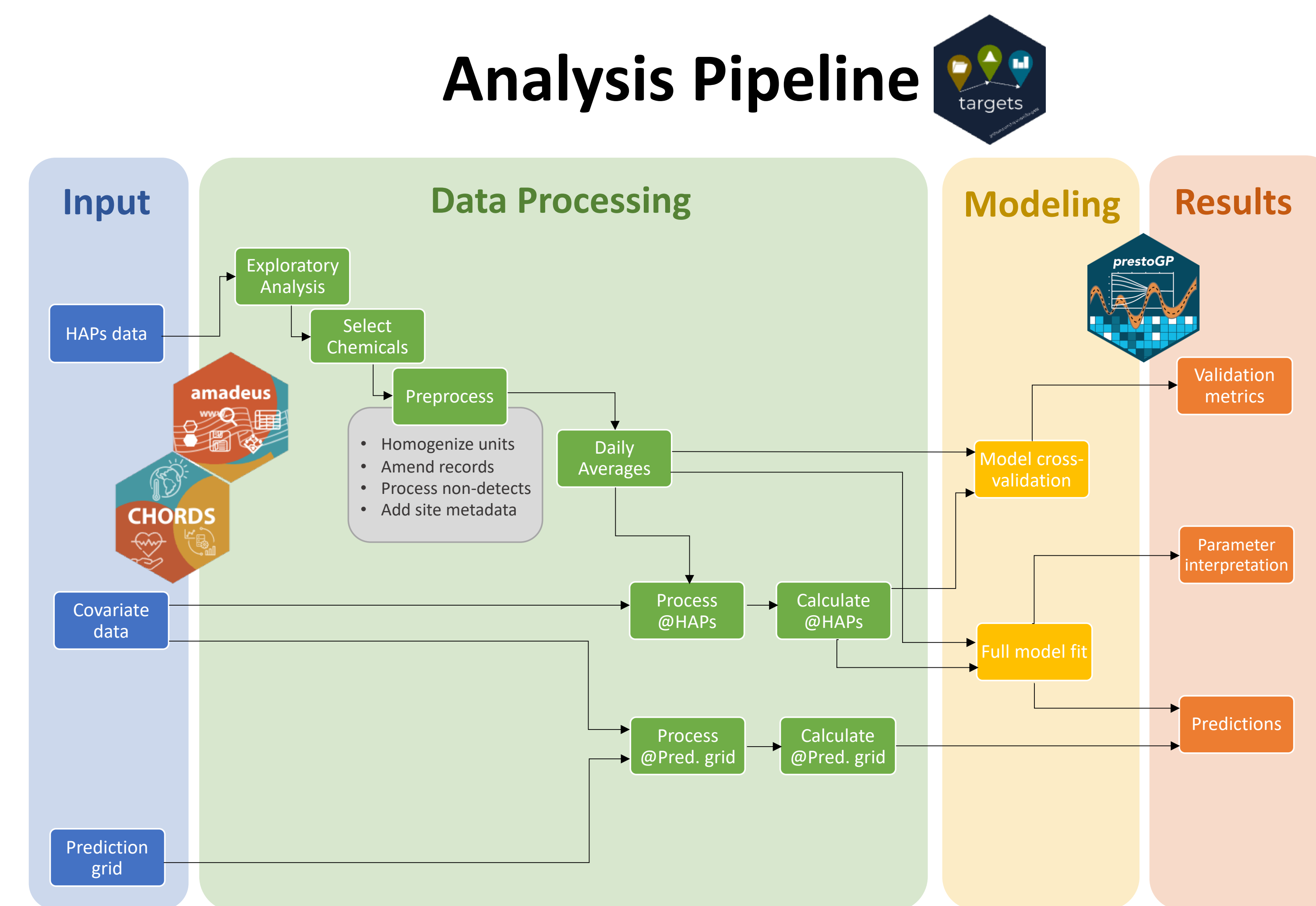


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Project Objectives

1. Map **HAPs concentrations** in the United States for recent years
2. Use **novel statistical and computational methods** for large datasets in multivariate scenarios and missing/censored data
3. Develop an **analysis pipeline** for easy reproducibility and expansion

Analysis Pipeline



Multivariate, Penalized, Land-Use Regression Model

Land-use regression between pollutants $\mathbf{z}$ and covariates $\mathbf{X}$:

$$\mathbf{z} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}_n(\mathbf{0}, \boldsymbol{\Sigma}_\theta)$$

Spatiotemporally correlated residuals

Expressed as Multivariate Gaussian density:

$$f(\mathbf{z}; \boldsymbol{\beta}, \boldsymbol{\theta}) = \mathcal{N}_n(\mathbf{z} | \mathbf{X}\boldsymbol{\beta}, \boldsymbol{\Sigma}_\theta)$$

To solve, we need to optimize the objective function:

$$Q(\boldsymbol{\beta}, \boldsymbol{\theta}) = \underbrace{-2 \log f(\mathbf{z}; \boldsymbol{\beta}, \boldsymbol{\theta})}_{\text{density}} + \underbrace{\lambda p(\boldsymbol{\beta})}_{\text{penalty}} = (\mathbf{z} - \mathbf{X}\boldsymbol{\beta})' \boldsymbol{\Sigma}_\theta^{-1} (\mathbf{z} - \mathbf{X}\boldsymbol{\beta}) + \log |\boldsymbol{\Sigma}_\theta| + \lambda p(\boldsymbol{\beta})$$

Problem: evaluation of $n \times n$ covariance matrix $\boldsymbol{\Sigma}_\theta$ takes $\mathcal{O}(n^3)$ time for each $\boldsymbol{\theta} \rightarrow$ computationally infeasible

Solution: use sparse approximation $\hat{Q}(\boldsymbol{\beta}, \boldsymbol{\theta})$ obtained through **Sparse General Vecchia Approximation** (Messier and Katzfuss, 2021)

Preliminary results

Parameters selected

Covariate	Benzene	Carbon Tetrachloride	Chloroform	Hexane
Max Temp	0.008		0.0005	0.029
Min Temp	-0.004			-0.028
Precip	0.002			0.003
Humidity	-38.38		-3.4361	-0.258
Radiation			0.0002	
Wind speed	-0.163		-0.0609	-0.191
Wind dir.	-0.0003			0.0000002

Covariates from Gridmet (~4km reanalysis product).

Future covariates: more meteorology, emissions inventories (TRI, NEI), land use (OpenLandMap), etc.

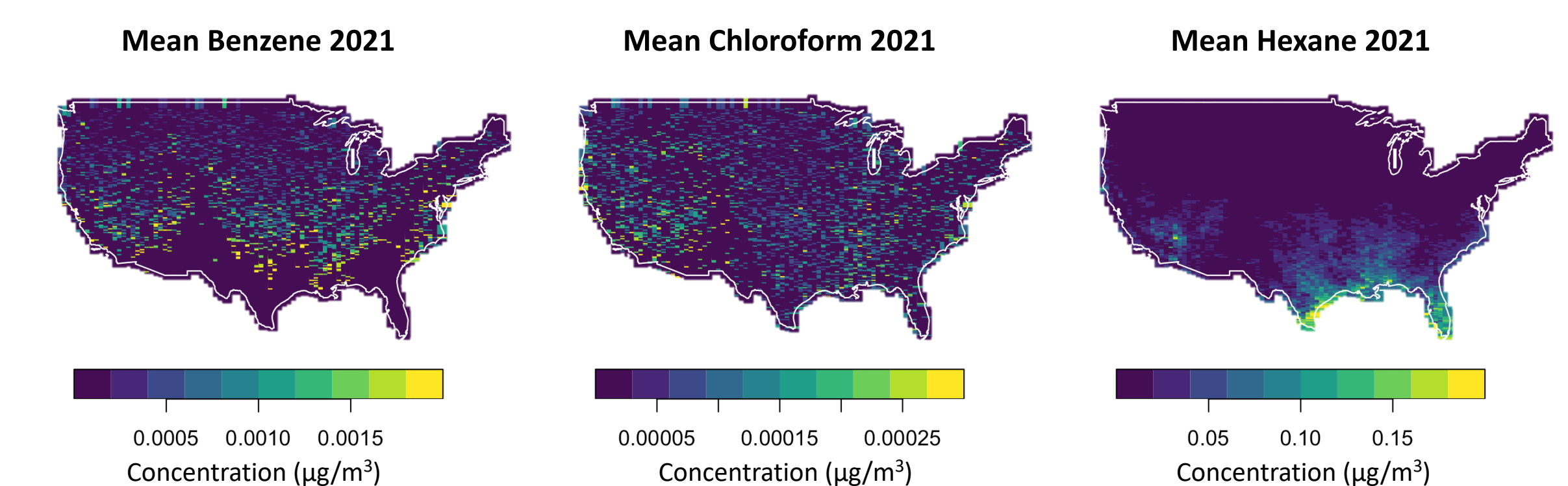
Validation metrics

Spatial block cross-validation (5-fold)

MSE results across all outcomes ($\mu\text{g}/\text{m}^3$): 5, 26, 0.8, 75, 2

Mean MSE across CV-folds: 22 $\mu\text{g}/\text{m}^3$

Prediction maps



Conclusions

1. We **modeled the concentrations** of four HAPs in the CONUS for 2021
2. We used a small initial set of data and covariates to **conceptualize the analysis pipeline**
3. Preliminary results show **areas for improvement**:
 - Heterogeneity in validation results $\rightarrow$ investigate data outliers and rethink cross-validation methods
 - Noisy model fit and predictions $\rightarrow$ addition of relevant covariates will allow for improvements across chemicals
4. Analysis pipeline development will **streamline the expansion** of this initial study, increasing computational efficiency and reducing user error

Future Work:

- Expanding to 2018-2021
- Implementing analysis pipeline
- Incorporating more covariates: meteorology, emissions, land use, seasons, etc.
- Health impacts and exposure disparities research